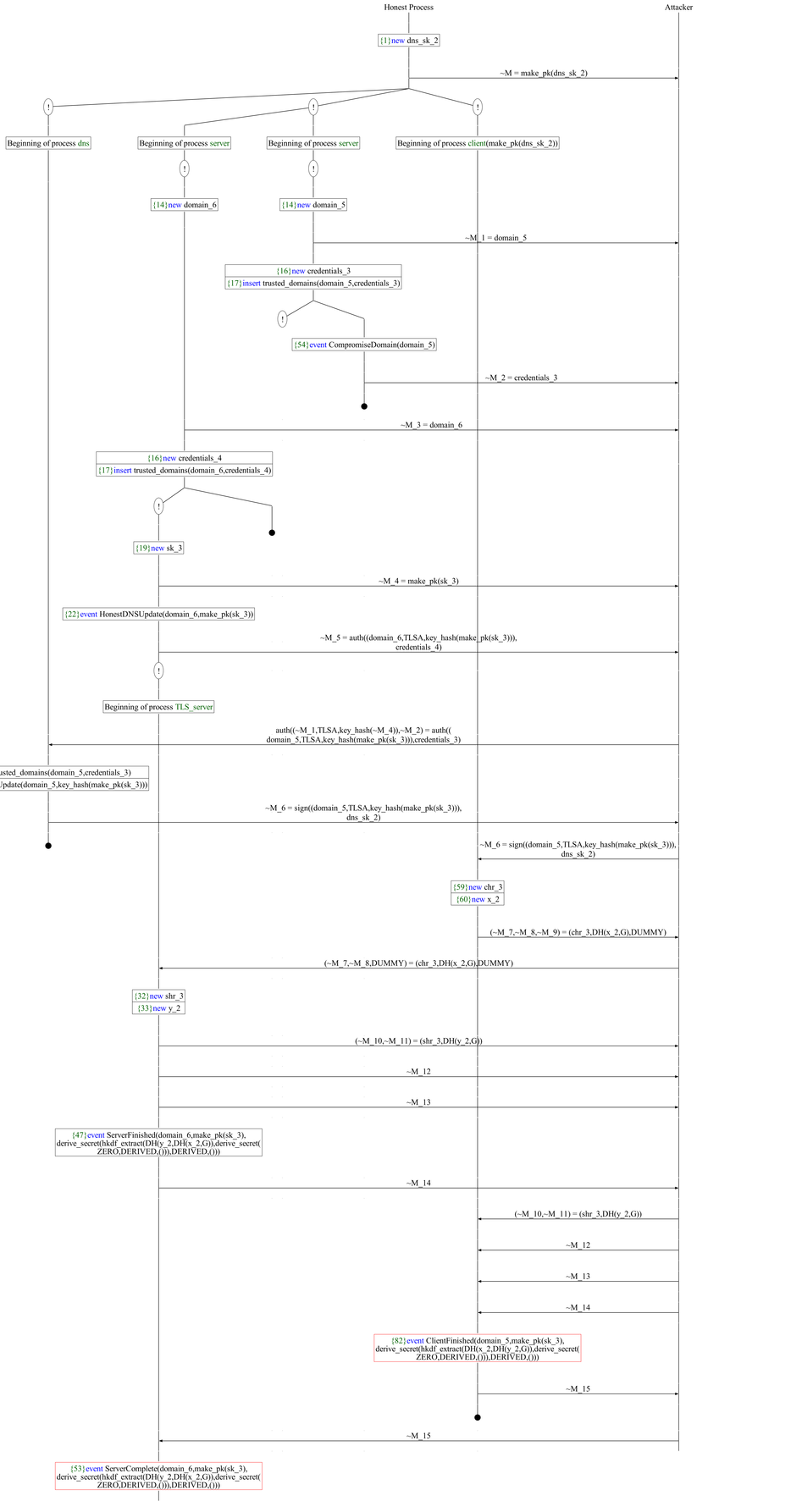
Abbreviations
~M_12 = enc((CERTIFICATE,make_pk(sk_3)),derive_secret(hkdf_extract(DH(y_2,DH(x_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G))))
~M_13 = enc((CERTIFICATE_VERIFY,sign(hash(((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)),enc((CERTIFICATE,make_pk(sk_3)),derive_secret(hkdf_extract(DH(y_2,DH(x_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),sk_3)),derive_secret(hkdf_extract(DH(y_2,DH(x_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G))))))
~M_14 = enc(FINISHED,mac(hash(((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)),enc((CERTIFICATE,make_pk(sk_3)),derive_secret(hkdf_extract(DH(y_2,DH(x_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),sk_3)),derive_secret(hkdf_extract(DH(y_2,DH(x_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),enc((CERTIFICATE_VERIFY,sign(hash(((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)),enc((CERTIFICATE,make_pk(sk_3)),derive_secret(hkdf_extract(DH(y_2,DH(x_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),sk_3)),derive_secret(hkdf_extract(DH(y_2,DH(x_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),derive_secret(hkdf_expand_label(derive_secret(hkdf_extract(DH(y_2,DH(x_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))),FINISHED)),derive_secret(hkdf_extract(DH(y_2,DH(x_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G))))))
~M_15 = enc(FINISHED,mac(hash(((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)),enc((CERTIFICATE,make_pk(sk_3)),derive_secret(hkdf_extract(DH(y_2,DH(x_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),sk_3)),derive_secret(hkdf_extract(DH(x_2,DH(y_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),enc((CERTIFICATE_VERIFY,sign(hash(((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)),enc((CERTIFICATE,make_pk(sk_3)),derive_secret(hkdf_extract(DH(y_2,DH(x_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),sk_3)),derive_secret(hkdf_extract(DH(x_2,DH(y_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),enc((FINISHED,mac(hash(((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)),enc((CERTIFICATE,make_pk(sk_3)),derive_secret(hkdf_extract(DH(x_2,DH(y_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),sk_3)),derive_secret(hkdf_extract(DH(x_2,DH(y_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),derive_secret(hkdf_expand_label(derive_secret(hkdf_extract(DH(x_2,DH(y_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))),FINISHED)),derive_secret(hkdf_extract(DH(x_2,DH(y_2,G)),derive_secret(ZERO,DERIVED,())),SHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))),derive_secret(hkdf_expand_label(derive_secret(hkdf_extract(DH(x_2,DH(y_2,G)),derive_secret(ZERO,DERIVED,())),CHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))),FINISHED)),derive_secret(hkdf_extract(DH(x_2,DH(y_2,G)),derive_secret(ZERO,DERIVED,())),CHTS,((chr_3,DH(x_2,G),DUMMY),(shr_3,DH(y_2,G)))))

A trace has been found.